



Machine Learning Lecture

What We Will Learn

1 Evaluation for Regression Models

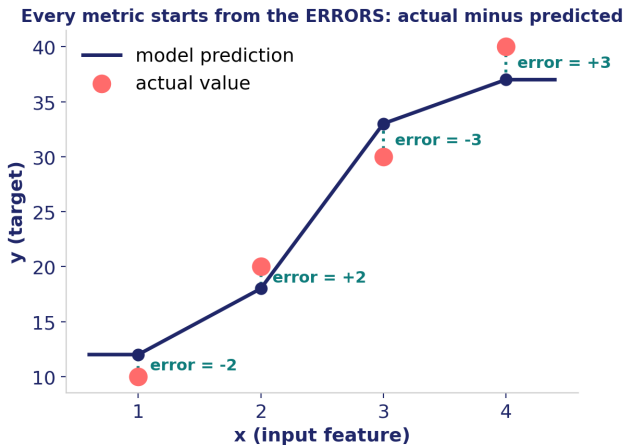
2 Evaluation for Classification Models

Our Running Example

A model predicts values. We compare prediction against reality:

Actual y	Predicted $\hat{y}$	Error $y - \hat{y}$
10	12	-2
20	18	+2
30	33	-3
40	37	+3

$n = 4$ points. Every metric is a different way of combining these 4 errors into ONE number. We build them one by one and explain WHY each was invented.



Step 0: Why Not Just Average the Errors?

The first idea: take the plain average of the errors.

Try it on our data

$$\text{Mean Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i) = \frac{(-2) + (+2) + (-3) + (+3)}{4} = \frac{0}{4} = 0$$

The result LIES: positive and negative errors cancel

Zero looks like a perfect model, but the model was wrong on every prediction. Error +3 and error -3 kill each other in the sum and the mistake disappears.

The lesson

Every error must be made POSITIVE before averaging, so mistakes cannot hide. Two ways to do that, two families of metrics: take the absolute value $|e_i| \rightarrow$ **MAE** • square it $e_i^2 \rightarrow$ **MSE and RMSE**

Metric 1: MAE (Mean Absolute Error)

Fix cancellation: take the absolute value of every error, then average.

Formula

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Calculate on our data:

$$\text{MAE} = \frac{1}{4} (|-2| + | +2| + |-3| + | +3|) = \frac{2 + 2 + 3 + 3}{4} = \frac{10}{4} = 2.5$$

Real meaning

On average the prediction misses by 2.5 units. Same units as y , so if y is in dollars, MAE is “off by 2.5 dollars on average.” Anyone can understand it.

- ▶ **Strength:** easy to explain; one crazy outlier cannot dominate it.
- ▶ **So why do we need anything else?** MAE has a blind spot shown next.

Why Does MSE Exist When We Already Have MAE? (Reason 1)

MAE's blind spot: it cannot tell these two models apart

	Errors	$\sum e_i $	MAE
Model A	2, 2, 2, 2	8	$8/4 = 2$
Model B	0, 0, 0, 8	8	$8/4 = 2$

Same MAE = 2! But Model B makes ONE catastrophic mistake. In real life (house prices, medicine doses) one giant error is far more dangerous than four small ones. MAE treats them as equal.

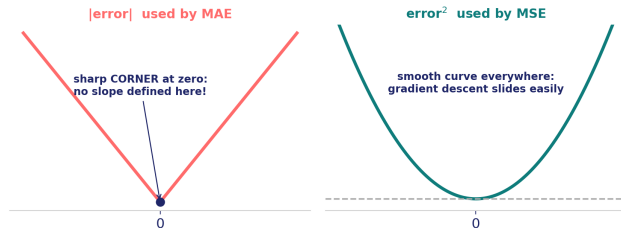
The fix: square every error before summing

Squaring grows fast: error 2 becomes 4, error 8 becomes 64.

$$\text{MSE}(A) = \frac{4 + 4 + 4 + 4}{4} = 4 \quad \text{MSE}(B) = \frac{0 + 0 + 0 + 64}{4} = 16$$

Model B is exposed. Big mistakes are punished heavily.

Why MSE? (Reason 2): No Derivative at Zero



- ▶ Training uses gradient descent: at every step we need the SLOPE (derivative) to decide which way to adjust the weights.
- ▶ $|e|$ has a sharp corner at zero: slope = -1 on the left, = $+1$ on the right, **undefined exactly at zero**. The optimizer gets confused right at the goal.
- ▶ e^2 is perfectly smooth: derivative = $2e$ everywhere, no corner, no confusion.

MSE wins twice

Reason 1: punishes big errors. Reason 2: smooth derivative for calculus. That is why **MSE is the standard training loss**, while MAE is used mostly as a reporting metric.

Metric 2: MSE (Mean Squared Error)

Formula

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Calculate on our data:

$$\text{MSE} = \frac{1}{4} ((-2)^2 + 2^2 + (-3)^2 + 3^2) = \frac{4 + 4 + 9 + 9}{4} = \frac{26}{4} = 6.5$$

- ▶ **Strength:** punishes big errors; smooth, used as training loss.
- ▶ **Weakness 1, units are squared:** y in dollars $\Rightarrow$ MSE in dollars². “Your error is 6.5 square dollars” is meaningless to a human.
- ▶ **Weakness 2, outliers dominate:** one point with error 100 adds 10 000 to the sum and drowns everything else.

New problem

MSE is great for the MACHINE (training) but bad for the HUMAN (reporting). We need the punishment of MSE, but back in readable units $\rightarrow$ that is RMSE.

Metric 3: RMSE (Root Mean Squared Error)

Formula

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Calculate: $\text{RMSE} = \sqrt{6.5} = 2.55$

The square root restores the original units; the heavy punishment of big errors survives inside.

Diagnostic trick: compare RMSE with MAE

RMSE is always $\geq$ MAE.

- ▶ $\text{RMSE} \approx \text{MAE}$ (ours: 2.55 vs 2.5): all errors are similar in size.
- ▶ $\text{RMSE} \gg \text{MAE}$: hidden BIG mistakes exist. Model B: $\text{MAE} = 2$ but $\text{RMSE} = \sqrt{16} = 4$, a warning sign!

Still open

Is 2.55 good or bad? For house prices it is amazing; for exam scores 0–10 it is terrible. RMSE has no universal scale $\rightarrow$ we need a unit-free score: R^2 .

Metric 4: R^2 (R-squared)

Formula: compare against the dumbest model (always predict $\bar{y}$)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\text{model error}}{\text{baseline error}}$$

Calculate: $\bar{y} = \frac{10 + 20 + 30 + 40}{4} = 25$

$$\sum_{i=1}^4 (y_i - \bar{y})^2 = 15^2 + 5^2 + 5^2 + 15^2 = 500 \quad \sum_{i=1}^4 (y_i - \hat{y}_i)^2 = 4 + 4 + 9 + 9 = 26$$

$$R^2 = 1 - \frac{26}{500} = 0.948$$

Real meaning

Our model explains 94.8% of the variation. $R^2 = 1$: perfect; $R^2 = 0$: no better than guessing the mean.
Unit-free and universal.

Why Adjusted R^2 When We Have R^2 ?

R^2 has a dangerous habit: it NEVER goes down

Add any new feature, even a useless one, and R^2 stays the same or increases. A model stuffed with junk features can LOOK better than a clean model.

Formula: charge a penalty for every extra feature

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \cdot \frac{n - 1}{n - p - 1} \quad n = \text{data points}, \quad p = \text{number of features}$$

Calculate with $R^2 = 0.948$, $n = 4$:

- ▶ $p = 1$ useful feature: $R_{\text{adj}}^2 = 1 - 0.052 \times \frac{3}{2} = 0.922$
- ▶ $p = 2$ (add one junk feature): R^2 stays 0.948, but $R_{\text{adj}}^2 = 1 - 0.052 \times \frac{3}{1} = 0.844$

Read the story

Plain R^2 hid the junk feature; adjusted R^2 dropped from 0.922 to 0.844 and exposed it. Use adjusted R^2 when comparing models with different feature counts.

The Whole Regression Story in One Table

Metric	Ours	Why it was invented	Weakness that led to the next
MAE	2.5	absolute value stops cancellation	blind to big errors, corner at zero
MSE	6.5	squares punish big errors, smooth	units squared, unreadable
RMSE	2.55	square root restores units	no universal scale: good or bad?
R^2	0.948	unit-free, 0–1, vs dumb model	never drops, junk features hide
Adj R^2	0.922	penalty per feature, honest	(the end of the chain)

One sentence per metric

MAE: how much do I miss on average. MSE: punish big misses and smooth for training. RMSE: the same punishment in human units. R^2 : how much of the data do I explain. Adjusted R^2 : the same, but honest about extra features.

First: The Confusion Matrix

Example: a spam filter checks 100 emails (60 spam, 40 normal).

	Model says: SPAM	Model says: NOT SPAM
Actually SPAM (60)	TP = 40 True Positive spam, model said spam CORRECT	FN = 20 False Negative spam, model said not spam MISSED!
Actually NOT SPAM (40)	FP = 10 False Positive not spam, model said spam FALSE ALARM!	TN = 30 True Negative not spam, model said not spam CORRECT

Reading the Four Boxes

- ▶ **TP (True Positive) = 40**: email was spam, model said spam. Correct catch.
- ▶ **TN (True Negative) = 30**: email was normal, model said normal. Correct pass.
- ▶ **FP (False Positive) = 10**: email was NORMAL but model said spam. A real email lands in the spam folder!
- ▶ **FN (False Negative) = 20**: email was SPAM but model said normal. Spam reaches the inbox!

Memory trick

The second letter is what the model SAID (P or N). The first letter says whether it was RIGHT (T) or WRONG (F).

Check the totals

$TP + FN = 60$ actual spam. $FP + TN = 40$ actual normal. All four = 100 emails.

Accuracy

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{\text{all correct}}{\text{all predictions}}$$

Calculate:

$$\text{Accuracy} = \frac{40 + 30}{100} = 0.70 = 70\%$$

Real meaning

The model judges 70 out of every 100 emails correctly.

When accuracy LIES: imbalanced data

In fraud detection only 1% of transactions are fraud. A lazy model that always says “not fraud” scores 99% accuracy and catches zero fraud. That is why we need precision and recall.

Precision: When the Model Says Positive, Can We Trust It?

Formula

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{true alarms}}{\text{all alarms raised}}$$

The model raised a spam alarm $40 + 10 = 50$ times, and was right 40 times:

$$\text{Precision} = \frac{40}{40 + 10} = \frac{40}{50} = 0.80 = 80\%$$

Real meaning

When the model shouts "SPAM!", it is right 80% of the time. 2 out of every 10 alarms are false alarms.

When precision matters most

When a **false alarm is expensive**: a false positive throws a real, possibly important email into the spam folder. We want high precision there.

Recall: Of All the Real Positives, How Many Did We Catch?

Formula

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{caught}}{\text{all that actually existed}}$$

There were $40 + 20 = 60$ real spam emails; the model caught 40:

$$\text{Recall} = \frac{40}{40 + 20} = \frac{40}{60} = 0.667 = 66.7\%$$

Real meaning

The model catches two thirds of the spam; one third slips through.

When recall matters most

When a **miss is expensive**: in cancer screening a false negative means telling a sick patient they are healthy. We want recall as high as possible, even if it costs some false alarms.

F1-Score: One Number for Both Precision and Recall

Formula (harmonic mean)

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

Calculate with Precision = 0.80 and Recall = 0.667:

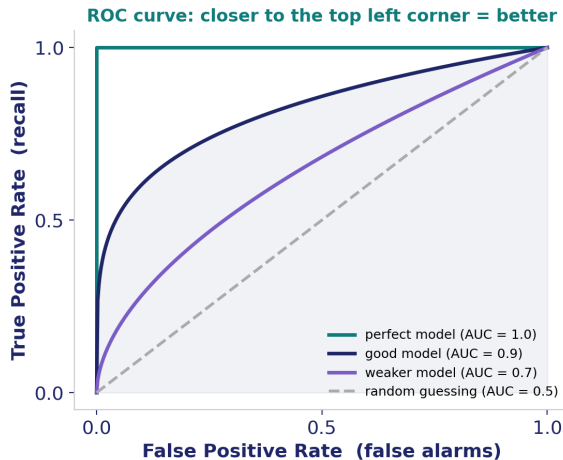
$$F_1 = 2 \times \frac{0.80 \times 0.667}{0.80 + 0.667} = 2 \times \frac{0.533}{1.467} = 0.727$$

Why the harmonic mean and not the plain average?

The harmonic mean punishes imbalance. If Precision = 1.0 but Recall = 0.0, the plain average says 0.5 (looks okay), but F_1 says 0 (the model is useless). F_1 is high only when BOTH precision and recall are high.

Use F_1 when data is imbalanced and you care about the positive class, which is most fraud, spam and disease detection problems.

ROC Curve and AUC



- ▶ A classifier outputs a probability (e.g. 0.83 spam). We choose a **threshold** to decide positive or negative.
- ▶ Every threshold gives one (TPR, FPR) pair. The **ROC curve** draws all of them.
- ▶ **AUC** = area under the curve, one number 0.5–1.

How to read it

Diagonal = random guessing (AUC 0.5).
Top-left corner = better. AUC 0.9: the model ranks a random spam above a random normal email 90% of the time.

Which Metric Should You Use?

Situation	Best metric	Why
Regression, easy to explain	MAE	average miss in real units
Regression, big errors are dangerous	RMSE	squaring punishes big misses
Compare models, different features	Adjusted R^2	penalizes junk features
Balanced classes	Accuracy	simple and fair
False alarms are costly	Precision	trust the alarms
Missing a positive is costly	Recall	catch them all
Imbalanced data, one number	F1-Score	needs both P and R high
Compare classifiers overall	ROC AUC	threshold independent

Summary

- ❶ **Plain mean error** = 0: positive and negative cancel. Every metric first makes errors positive (absolute value or squaring).
- ❷ $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| = 2.5$: average miss, readable.
- ❸ $\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 6.5$: punishes big errors and smooth for calculus, but units are squared.
- ❹ $\text{RMSE} = \sqrt{\text{MSE}} = 2.55$: readable units; $\text{RMSE} \gg \text{MAE}$ warns of big hidden mistakes.
- ❺ $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} = 0.948$: unit-free, how much variation is explained.
- ❻ Adjusted $R^2 = 0.922$: subtracts a penalty for extra features.
- ❼ Confusion matrix: TP 40, TN 30, FP 10, FN 20. Accuracy 70%, Precision 80%, Recall 66.7%, $F_1 = 0.727$.
- ❽ ROC AUC: overall comparison across all thresholds.